



SIGNAL ANALYSIS REPORT

Automated Signal Analysis and Parameter Extraction

SIH Signal Analyzer • National Technical Research Organisation

Date / Time: 2026-09-26 23:05:22

Input File: synthetic_QPSK_1000kHz_SNR20dB.wav

Format: WAV Audio / IQ (.WAV)

1. FILE / SIGNAL INFORMATION

File Name	synthetic_QPSK_1000kHz_SNR20dB.wav
File Format	WAV Audio / IQ (.WAV)
File Size	67,628 bytes
Signal Duration	16.90 ms (0.01690 s)
Sample Rate (Fs)	1.0000 MHz (1,000,000.0 Hz)
Center Freq Offset	883.19 Hz
Total Samples	16,896

2. DETECTED SIGNAL PARAMETERS

Modulation	QPSK
Modulation Confidence	100.00%
Occupied Bandwidth	174.805 kHz (174,804.7 Hz)
Carrier Freq Offset (CFO)	883.19 Hz
Symbol Rate	125,000.0 symbols/sec
Samples Per Symbol	8 sps
Frame Sync Status	Locked

3. MODULATION CLASSIFICATION

Modulation Scheme	Probability
QPSK	100.00%
GMSK	0.00%
BPSK	0.00%

4. TIME-DOMAIN WAVEFORM

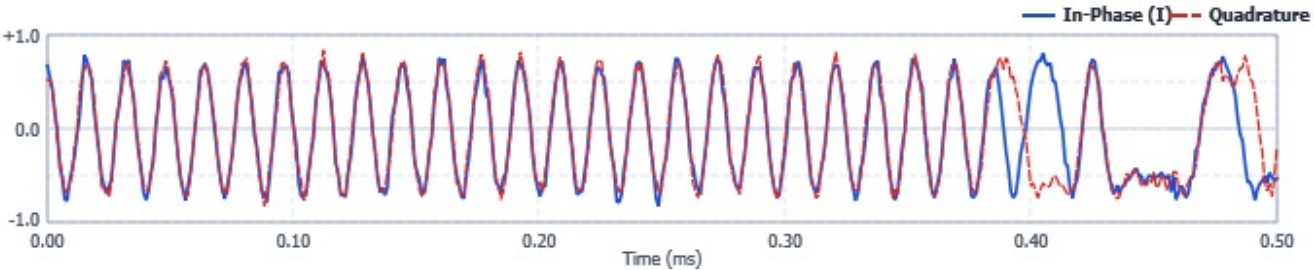


Figure 1: Time-Domain Waveform (In-Phase I & Quadrature Q Components)

5. FREQUENCY SPECTRUM

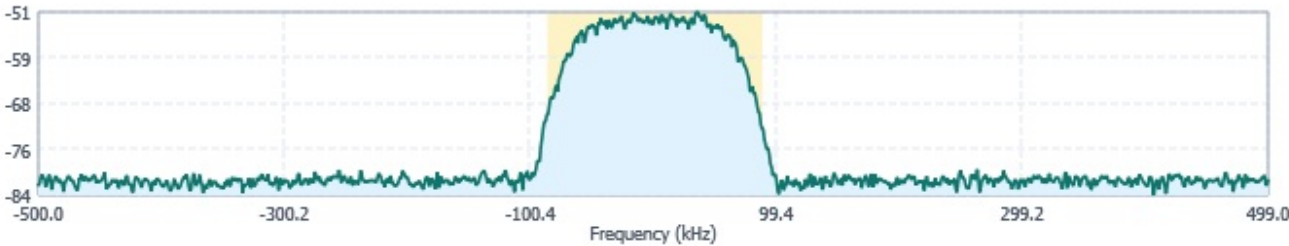


Figure 2: Power Spectral Density (PSD) and Occupied Bandwidth

6. SPECTROGRAM / WATERFALL

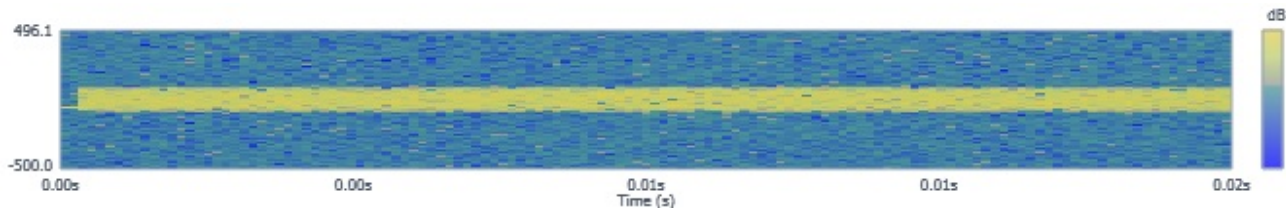
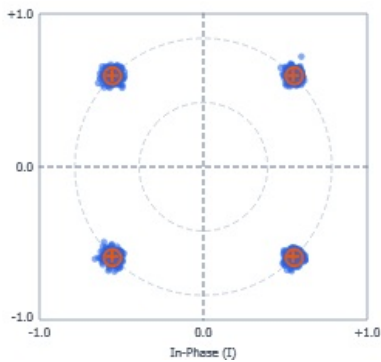


Figure 3: Time-Frequency STFT Spectrogram Waterfall

7. CONSTELLATION DIAGRAM



8. SYNCHRONIZATION / FRAME ANALYSIS

Frame Sync Status	Locked
Header Offset	124 bits
Payload Offset	156 bits
Sync Pattern Locked	CCSDS_ASM
Correlation Sync Match	100.0%
Phase Ambiguity Resolved	No (Direct polarity)
Demodulated Bits	3,910 bits

Figure 4: IQ Constellation

9. BITSTREAM / DECODING

Total Decoded Bits	3,754 bits	Decoded Bytes	470 bytes
Bit Density (1s / 0s)	1,888 (50.3%) / 1,866 (49.7%)	ASCII Text Preview	..+....}Rp..S.>.+..g...0.~@...[./.: \$.6d.rI.j5.)h..u....J.r.....?fW
Bitstream Sample	10101101 11110000 00101011 11000111 11101000 10111011 00010011 01111101 01010010 01110000 00011110 11100001 ... (+3658 bits)		

Decoded Bytes Hex Dump:

000000	AD F0 2B C7 E8 BB 13 7D 52 70 1E E1 53 C7 3E 00	..+....}Rp..S.>.
000010	2B D8 7F 67 89 1A 85 4F B1 7E 40 04 F4 5B DF 2F	+..g...0.~@...[./
... (438 additional bytes omitted)		

10. FEC / DE-INTERLEAVER ANALYSIS

Component	Detected Configuration	Status / Confidence
De-interleaver	none	Bypassed
Forward Error Correction (FEC)	none	Uncoded
Channel Decoder	Automated Multi-Scheme Pipeline	470 bytes recovered

11. FINAL ANALYSIS SUMMARY

RF capture file 'synthetic_QPSK_1000kHz_SNR20dB.wav' was processed, containing 16,896 samples recorded at a sample rate of 1.0000 MHz (1,000,000.0 Hz) (effective duration: 16.90 ms (0.01690 s)). The neural modulation classifier identified the transmission format as QPSK with a classification confidence of 100.00%. Spectral analysis determined an occupied bandwidth of 174.805 kHz (174,804.7 Hz) with a carrier frequency offset of 883.19 Hz. Demodulation recovered 3,910 raw bit transitions. Frame synchronization locked to CCSDS_ASM (100% match) with header boundary at bit 124 and payload commencing at bit 156. Channel decoding detected direct (no) interleaving, and raw uncoded bit framing, successfully assembling 470 decoded payload bytes.